

Spin vs. Substance: Trading Biotech Trial Readouts

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github.com/JBailey0703/gator-quant-hacks

1. Summary

We trade US small- and mid-cap biotech stocks after clinical-trial press releases. Google Gemini reads each release next to the primary endpoint the company registered on ClinicalTrials.gov *before* the release (when the trial can be matched) and scores its **spin**: how far the message drifts from what the trial was registered to test. We short spun releases that the market believed and follow clean results, holding each position 60 trading days. The edge we hypothesize is behavioral and structural: investors price the headline, and costly short selling slows the correction. Run once on frozen rules, the **out-of-sample test (Jan 2025 – Oct 2026, 107 trades) returned +22.9% a year net of costs, with a Sharpe ratio of 1.52 (95% interval 0.03–3.01) and a maximum drawdown of –11.9%**, during a biotech rally (XBI +68.7%). Replacing Gemini's score with a keyword count lowers out-of-sample Sharpe to 0.81. The evidence is promising, not conclusive: alpha after the XBI biotech index and the S&P 500 is +12.5% a year but not statistically significant ($t = 1.5$), the 2024 validation check failed, and a few large clean-success longs earned most of the profit.

2. Economic Hypothesis

Committed before any results: liquid small- and mid-cap biotech stocks move toward what the trial data actually support over the 5 to 20 trading days after a trial-results press release, because investors price the headline, not the substance. A spun release pushes the price too high and it falls back; a clean result is underpriced at first and keeps drifting in its direction.

Who is on the other side. Headline-driven investors, largely retail traders, who are net buyers of attention-grabbing stocks [1], in small biotechs that few analysts cover. Spin in trial reporting is well documented: it is common when a trial's primary outcome fails [2], and about half of the press releases in one cohort study contained it [3].

Why the opportunity persists. *Behavioral bias (primary):* spin invites overreaction to the headline, and checking a release against the endpoint registered months earlier takes expertise and time that few investors spend, so the

substance reaches prices slowly [4], much like post-earnings-announcement drift [5]. That check is the step we automate with Gemini. *Structural constraint (primary):* shorting small biotechs is costly and risky (borrow fees, recalls, squeezes), so optimists set prices [6] and arbitrage capital is limited [7]. *Risk premium (secondary):* part of the long return may pay for company-specific binary event risk, which a low market beta does not rule out. *Liquidity provision (minor):* we trade at the close a day after the news, so we take liquidity rather than supply it, and target slow interpretation, not a market maker's spread.

Predictions and failure conditions. (1) Higher spin leads to a larger decline; (2) clean results keep drifting; both within 5–20 trading days. The strategy fails if high- and low-spin releases perform alike, clean results show no drift, or the out-of-sample result is near zero after costs.

3. Data & Universe

Instruments, frequency and dates. US-listed pharmaceutical and biotech stocks (SEC SIC 2834, 2836, 8731), long and short, on daily closes from May 1, 2018 to Oct 2, 2026. Before each reaction day a stock needs a market cap of \$300M–\$10B (latest shares filed $\times$ prior close) and a 20-day average Nasdaq dollar volume of at least \$1.2M. Benchmark: XBI. Periods: *in-sample* May 2018 – Dec 2023 (rules and holding period chosen), *validation* 2024 (confirmation only), *out-of-sample* Jan 2025 – Oct 2026 (run once). An event counts in a period only if its full 60-day hold ends inside it.

Sources. SEC EDGAR full-text search, filings and XBRL [8]; ClinicalTrials.gov [9] and the AACT monthly archives [10], [11] (trial matching and endpoints as registered before each release, 35 snapshots); Databento XNAS.ITCH hour bars, including delisted stocks [12]; Google Gemini 3.1 Flash-Lite [13]; Massive 8-K tags for a recall check of our search (88%) [14]; and Webull OpenAPI closes [15], via the hackathon starter code [16], as a price cross-check.

Event timing. Event times are SEC acceptance timestamps in New York time, a proxy for when the release became public. Releases with only a

date are assumed to come after the close, so no reaction is measured before the news; releases dated after their filing, or filed over 7 days late, are dropped.

Survivorship, corporate actions, missing data.

Delisted stocks are included where Databento has their history, and we use the ticker printed in the release rather than today's ticker lists; this reduces, but does not remove, survivorship and ticker-mapping risk. Returns use unadjusted closes and ignore dividends. A one-day rise above $3\times$ on low volume, or with a reverse-split 8-K, has its split ratio removed (no trade was held through one). A day without trades counts as a zero return, and events missing a close on the previous, reaction or entry day are dropped. A stock that stops trading is exited at its last close (3 of 376 in-sample and validation trades, 2 of 107 out-of-sample); side results where those longs lose 30% or 100% bracket published delisting returns of -30% (NYSE/AMEX) [17] and -55% (Nasdaq) [18].

4. Methodology

Measuring spin with Gemini. Gemini 3.1 Flash-Lite is our measuring instrument, never the trader: a fixed rule (Eq. 2) makes every decision. Comparing a release with its registered endpoint is what a keyword list cannot do, since a release can mislead using only positive words, for example by never mentioning the endpoint that failed. Gemini works at five steps (Table 1). Every call uses temperature 0, a fixed JSON answer format and no web search, so the model sees only the text we send, and all 28,336 answers are saved in the repository, so the backtest re-runs without new calls. The prompts include four of our 15 hand-labeled releases as examples; the rest are held out.

Table 1. Where Gemini works in the pipeline

Step	Question for Gemini	Answers
Filter	new human trial results?	13,739*
Extract	ticker, drug, phase, repeat?	3,702
Match	which trial is reported?	1,950
Check	same drug and disease?	2,027
Score	spin 0–4; endpoint met?	6,918†

* After a keyword rule removed 12,094 earnings releases. † Three scoring versions per readout.

From filings to trades. Of 25,833 candidate filings, Gemini kept 3,702 announcing new human trial results; removing repeats and contradictory dates left 2,833 readouts; 800 passed the universe, price and full-window filters; 483 became trades.

Signal construction. For readout i , Gemini (G) reads the release with company names, tickers, dates, people and drug codes redacted ($\tilde{D}_i$) next to R_i , the matched trial's endpoints and design in the latest AACT snapshot before the release (a median 52 days earlier), never its status, results or enrollment:

$$(S_i, M_i) = G(\tilde{D}_i, R_i) \quad (1)$$

Without an accepted match or an earlier snapshot, R_i is empty and the release is scored on its own (984 of 2,833 readouts). Gemini returns whether the primary endpoint was met, $M_i \in \{\text{yes, no, unclear}\}$, and spin $S_i \in \{0, \dots, 4\}$ on a rubric built from our hand labels: 0 accurate, 1 mild promotion, 2 meaningful reframing via secondary endpoints or subgroups, 3 substantial reinterpretation, 4 a failed endpoint presented as success.

Table 2. Checks on Gemini's measurements

Check	n Agree	
Endpoint met vs held-out hand labels	6	6
Spin within 1 point of hand labels	6	5
Trial match correct (audit)	30	28
Same clean/spun group as audit	30	26
Spin within 1 point of audit	30	30
Rejected matches truly wrong	15	≈ 11

The 30 audited events were labeled blind by an AI reader, then checked by a team member: a team check, not an independent study. All four group disagreements were borderline 1-versus-2 calls. The match check was revised after the audit found its two wrong matches; 15 later rejections test it fresh.

From score to signal. Let AR_i be the stock's close-to-close return minus XBI's on the first session whose close follows the release. The position direction is

$$d_i = \begin{cases} -1 & S_i \geq 2 \text{ and } AR_i > 0 \\ +1 & S_i \leq 1 \text{ and } M_i = \text{yes} \\ -1 & S_i \leq 1 \text{ and } M_i = \text{no} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

The cutoffs come from the rubric, fixed before any results. A clean release with an unclear endpoint (28% of registry-scored readouts) is not traded. The signal is final at the reaction-day close and the trade fills at the next session's close, so no signal uses the close it trades at.

Portfolio, sizing and rebalancing. Starting capital is \$1M. Each position is sized once, at entry:

$$x_i = \min \left(0.05E_t, 0.01V_i, \frac{E_t - A_t}{1 + c} \right) \quad (3)$$

where E_t is equity, V_i the stock's 20-day average Nasdaq dollar volume, A_t the value of positions already open and $c = 0.2\%$ the fee; there is no trade if $x_i \leq 0$. At most 20 positions, one per stock; earlier announcements get scarce slots first. There is no rebalancing: positions are marked to market daily and exit after 60 trading days, or at the last close if trading stops. Execution: fills modeled at the regular-hours close (the last Nasdaq trade), 20 bps per side, and 10% a year to borrow shorted shares.

Holding period and discipline. A pre-registered rule picked the holding period on in-sample data only: among 1, 2, 3, 5, 10, 20, 40 and 60 days, the longest within 0.05 of the best net Sharpe whose neighbors on that list were also profitable. It chose 60, the longest horizon pre-specified; at 40 days the Sharpe was 0.34 against 0.67, so 60 is a peak, not a broad plateau. The 2024 confirmation failed (Sharpe -0.61) and, as the rules required, the out-of-sample run used 60 days anyway. In total we report one strategy, 8 horizons, 12 comparison variants and 9 parameter neighbors, none selected after the fact. The hypothesis, every rule change and the out-of-sample lock (a fingerprint of all inputs and code) are time-stamped in the repository. One command, `run_all.py`, re-runs the backtest (needs a Databento key and about \$8 of data) and checks every result table against the committed ones; `run_all.py --check` verifies the headline numbers with no keys.

5. Results

Out-of-sample, the strategy earned +22.9% a year net of costs (Sharpe 1.52). That period was a biotech rally (XBI +68.7%). Buying XBI earned more (+35.1% a year) with about twice the volatility, so the strategy's Sharpe was higher (1.52 vs 1.20). Its beta to XBI was only 0.30, and a regression on XBI and the S&P 500 leaves alpha of +12.5% a year (information ratio 1.13) that is not statistically significant ($t = 1.5$), so we do not claim proven skill.

Table 3. Performance, net of all costs: in-sample 2018–23, validation 2024, out-of-sample 2025–26

	2018–23	2024	2025–26
Annual return	+8.8%	−5.3%	+22.9%
Volatility	14.0%	8.4%	14.2%
Sharpe ratio	0.67	−0.61	1.52
Max drawdown	−19.2%	−10.0%	−11.9%
Turnover per year	3.7×	3.9×	4.5×
Trades	325	51	107
Win rate	48%	37%	55%
Beta to XBI	0.27	0.15	0.30
Alpha per year	+7.2%	−8.1%	+12.5%
Alpha t-statistic	1.6	−1.1	1.5

Beta: regression on XBI alone. Alpha: regression on XBI and the S&P 500. Turnover: traded value, both sides, over average equity.

In-sample and validation. In-sample, the strategy earned +8.8% a year (Sharpe 0.67), mostly in 2019–2021; 2024 lost 5.3%. Sharpe 95% intervals, assuming independent daily returns [19], are -0.16 to 1.49 in-sample, -2.57 to 1.36 in 2024 and 0.03 to 3.01 out-of-sample: only the out-of-sample Sharpe is distinguishable from zero, and only just ($t = 2.0$). Overlapping positions make these intervals approximate.

Where the profit comes from. Clean-success longs earned almost all of it (Table 4): \$425k of \$431k out-of-sample. The five best trades were 79% of in-sample P&L (Novavax alone, +419% in 2020) and 61% out-of-sample; leaving them out of the tally still leaves +\$129k and +\$166k on \$1M. Spin matters for the longs too: the long side earned more than in the variant that ignores spin in all three periods (+\$20k, +\$22k and +\$73k), consistent with spin keeping weak "successes" out, though the two portfolios hold different stocks. Spun shorts were profitable in-sample and out-of-sample, but small; clean-failure shorts lost money in two of three periods.

Table 4. Net P&L by trade type (\$ thousands)

Trade type	2018–23	2024	2025–26
Clean success → long	+547	−60	+425
Spun, rose → short	+66	−10	+24
Clean failure → short	−6	+16	−17
Total	+608	−53	+431

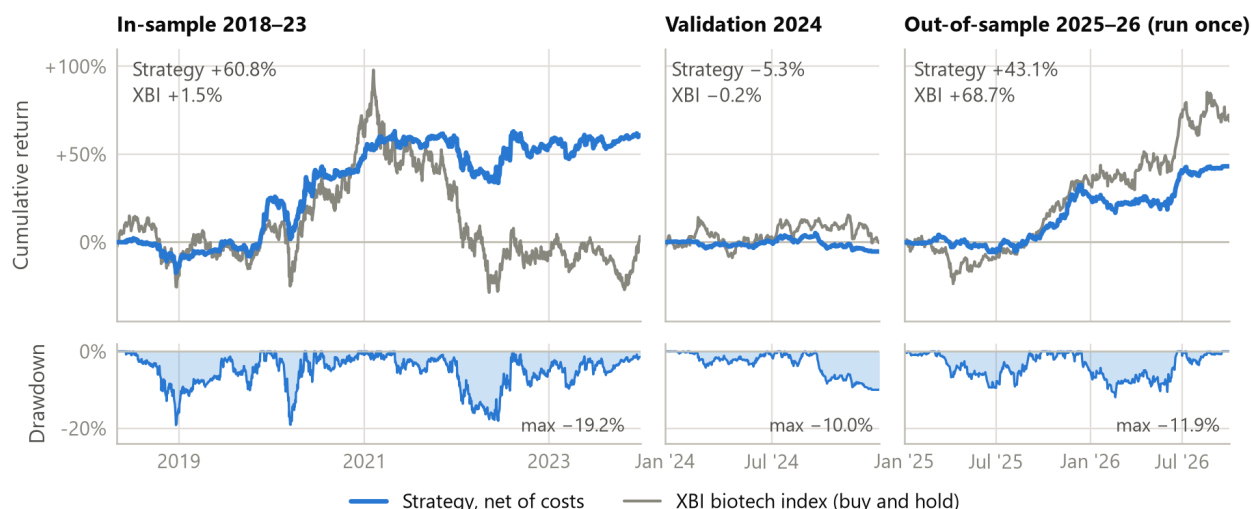


Figure 1. Cumulative return in each period, net of costs, against buy-and-hold XBI over the same days; each period is simulated separately from \$1M. Drawdowns below.

What each ingredient adds. Figure 2 tests our central idea one change at a time (differences not tested for significance). Replacing Gemini's spin score with a ten-pharse keyword count lowers Sharpe in all three periods (0.15 / 0.81 vs 0.67 / 1.52 in-sample / out-of-sample), and ignoring spin lowers it too (0.46 / 1.09). Out-of-sample, scoring against the pre-release registry beat both the release alone (1.21) and today's registry record (0.88), though not in-sample. Fading every day-one rise without spin loses money, so the result is not a simple reversal effect. Results also survive doubled costs (0.53 / 1.38), long-only trading (0.49 / 1.18) and a 100% loss on delisted longs (0.46 / 0.98).

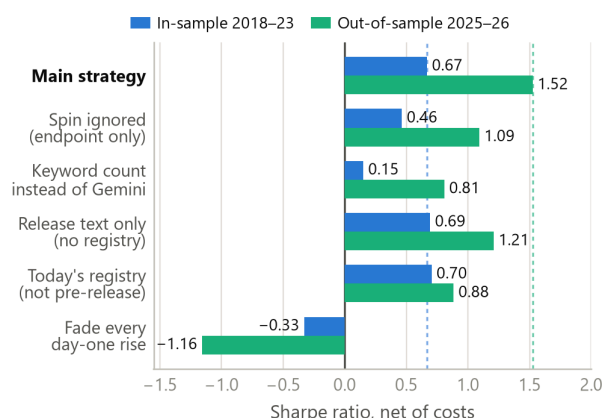


Figure 2. Net Sharpe ratio when one ingredient changes; dashed lines mark the main strategy. In 2024 every row was negative (−0.19 to −2.38).

Where the edge lives. Across all eligible events (Appendix Fig. A2), out-of-sample, spun releases

that rose on day one went on to fall 12.0% against XBI over 60 trading days (95% interval −23.5% to +0.5%, 18 events): the reversal our hypothesis predicts, but the interval includes zero. Clean successes gained 3.2% (−5.5% to +15.4%, 109 events). In-sample, every group's 60-day interval includes zero, and spin shows no steady pattern from 0 to 4, so prediction 1 is not supported. Timing also differed: one-day holds lose money in every period and out-of-sample Sharpe rises with holding time (Appendix Fig. A1), so the 60-day hold did best and the 5–20 days we predicted is not supported.

Nearby parameters. A sensitivity grid declared before it was run (after the out-of-sample test; Appendix Table A1) keeps every neighbor profitable in-sample (Sharpe 0.25–0.82) and out-of-sample (0.78–1.76), while 2024 loses under every setting (−0.18 to −0.94), so the 2024 failure is not a parameter artifact. The rubric's spin cutoff beats both neighboring cutoffs in-sample and out-of-sample.

6. Risk Management

Position and exposure limits. Limits apply at entry (Eq. 3) and there is no leverage. Without rebalancing, winners can drift: the largest position reached 19.8% of equity (Novavax, 2020), and one exceeded 10% on 2% of in-sample days. The book ran 21–29% net long on average.

Stop and de-risking rules. Each position has a 60-day time exit (5 left earlier because trading stopped), and new signals are skipped once 20 positions are open. There are no price stops:

biotech news arrives as overnight gaps, so a failed trial can cut a stock by half or more before the open, far through any stop. The cost is open-ended short risk: the worst trade, a 2020 short in HGEN, lost 162% of its entry size (1.6% of equity); the worst long lost 93% (FGEN, 2023).

Table 5. Risk controls in the backtest

Control	Limit	Observed
Per stock	5% at entry	max 19.8%
Liquidity	1% of V_i	binds 53–65%
Positions	20, one per stock	avg. 12–14
Gross	100% at entry	max 79–92%
Time exit	60 days	5 earlier
Price stop	none	worst –162%

Factor and correlation exposure. Beta to XBI is 0.27 / 0.15 / 0.30 and to the S&P 500 0.33 / 0.25 / 0.17 (in-sample / 2024 / out-of-sample). Stocks held at the same time are moderately correlated (0.21–0.30), but because the book holds longs and shorts, the P&L on those positions is nearly uncorrelated on average (0.01–0.09); an average can still hide losses that arrive together.

Tail and regime risk. The worst day was –3.8% (Mar 16, 2020), the worst month –16.8%, and a 1-in-100 day loses 1.3–2.4%; returns are fat-tailed (excess kurtosis up to 4.4) but mostly right-skewed. In biotech's five worst in-sample months (XBI –15.4% on average) the strategy averaged –5.4%. Live, we would add three rules (untested, to avoid tuning after the results): pause new entries after a 20% drawdown, trim positions above 10% of equity, and short only with confirmed borrow.

7. Liquidity & Capacity

Trading costs and slippage. We charge 20 bps per side and 10% a year on shorts, with turnover of 3.7–4.5× a year; doubling both leaves Sharpe at 0.53 in-sample and 1.38 out-of-sample. We model fills at the last Nasdaq trade before the close, a median 0.01–0.02% from Webull's official close for the two stocks checked; that confirms the price, not that an order would fill there, so the 20 bps also stands in for slippage.

Market impact and capacity. The median position is about \$45k in companies with a median market cap of \$1.6–1.8B, and the volume cap was the binding limit for 53–65% of positions. For the capacity test each trade also pays square-root impact [20] on entry and exit:

$$I_i = \sigma_i \sqrt{\frac{x_i}{V_i / 0.24}} \quad (4)$$

where I_i is the cost as a fraction of the trade, σ_i the stock's 20-day daily volatility, and 0.24 Nasdaq's share of total volume, measured against Webull consolidated volume for two large biotechs. Scaling capital in-sample, annual return falls from 7.9% at \$1M to 5.3% at \$5M, 3.2% at \$10M, 1.4% at \$25M and 0.4% at \$100M, almost unchanged with a 15% or 35% Nasdaq share. Extra capital mostly sits idle under the volume cap, so **returns erode beyond roughly \$5–10M**: this is a small-capacity strategy (an in-sample estimate; capacity was not tested out-of-sample).

8. Limitations & Next Steps

What didn't work. The 2024 validation lost money under every setting; more spin did not mean a larger decline; clean-failure shorts did not pay; the 60-day hold beat the 5–20 days we predicted; and the spun-short reversal is not statistically significant.

What could break the strategy. A biotech bear market would hurt a book that runs net long. Crowding, as language models make this kind of reading cheap, or companies changing how they word releases could erase the signal, and prices often move before trial announcements [21]. Shorts depend on borrow that may be unavailable or recalled, which a constant 10% fee ignores. Profits rest on a few winners and one 21-month rally.

Known approximations. Split ratios are inferred; exits after trading stops assume a fill at the last close; the 9:00 hour bar adds some pre-market volume to liquidity figures; drug-and-disease matching can confuse two cohorts of one drug; and registry snapshots are a median 52 days old (max 125). Our accuracy checks are small and run by our own team. Gemini may recognize in-sample events despite redaction (trial IDs are not redacted); its knowledge cutoff is January 2025 [13], so only the 32 out-of-sample readouts from that month could be in its training.

What we would test with more time. A few hundred independently labeled releases; an event timeline (medical meetings, publications) for the 60-day horizon; monthly registry snapshots; an XBI-hedged version; real borrow-fee data; the live rules above, on new data only; and Webull paper trading with the same daily pipeline.

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Appendix A. Supplementary figures and table

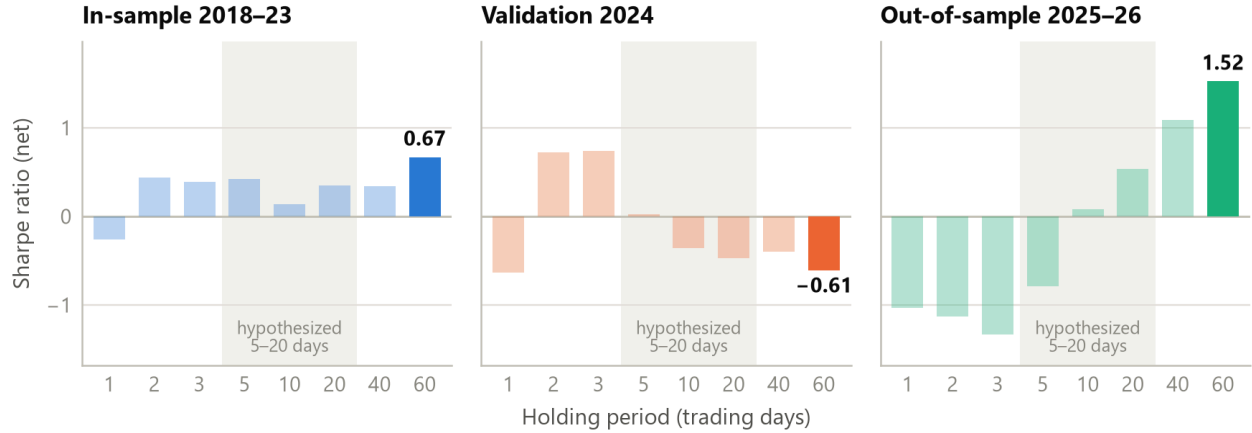


Figure A1. Net Sharpe ratio by holding period; the darker bar is the 60-day horizon chosen by the pre-registered rule on in-sample data and then fixed for 2024 and the out-of-sample run.

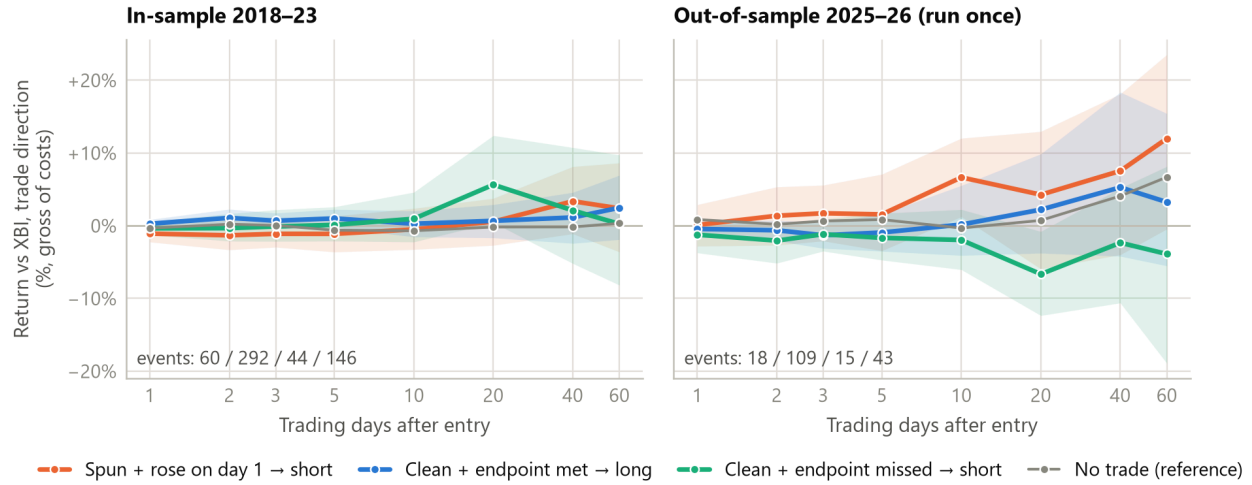


Figure A2. Mean return of all eligible events against XBI, in each trade's direction (a short gains when the stock falls), before costs and ending early if a stock stops trading; shaded: 95% intervals resampling companies. Event counts: spun-short / clean-long / clean-short / no trade.

Table A1. Parameter sensitivity, one change at a time (net Sharpe)

Setting	2018-23	2024	2025-26
Main strategy	0.67	-0.61	1.52
Spun = spin ≥ 1	0.25	-0.31	1.19
Spun = spin ≥ 3	0.45	-0.47	1.13
Short if day one $> 2\%$	0.62	-0.46	1.52
Max 3% per stock	0.53	-0.60	1.66
Max 10% per stock	0.80	-0.79	1.68
Volume cap 0.5%	0.82	-0.57	1.56
Volume cap 2%	0.47	-0.64	1.76
Max 10 positions	0.76	-0.94	0.78
Max 30 positions	0.60	-0.18	1.25